Baseline Models and Proposed Graph Neural Network

— GNN-Based Gesture Recognition on NinaPro DB1

Group04 — CSE475

Task 02 Report

Abstract

Building on the Task 1 EDA, this report establishes classical and deep-learning baselines for NinaPro DB1 gesture classification under a leakage-free, subject-independent protocol, and proposes a correlation-graph GCN as an initial GNN model. Nine baselines were evaluated (Logistic Regression, SVM, k-NN, Decision Tree, Random Forest, MLP, XGBoost, LightGBM, 1D-CNN); SVM (RBF kernel) is the strongest, with Accuracy 0.7746 and Macro-F1 0.7842 on a held-out set of subjects 21–27 (trained on subjects 1–20). The proposed GNN — 32
channel-nodes (EMG+glove) connected by a training-set correlation graph, refined by a learnable edge-weighting layer, through three GCN layers — substantially underperforms every classical baseline, achieving Macro-F1 0.3925, below even the Decision Tree baseline (0.4715). This result is reported directly, motivating the Task 3 ablation study.

Index Terms — sEMG, gesture recognition, subject-independent evaluation, GCN, baseline comparison

I. Introduction

Task 1 established that NinaPro DB1's 32 EMG+glove channels are windowed into 30,936 labeled examples across 23 classes, with known feature redundancy, class imbalance, and a cross-exercise label collision. This report uses that same windowed representation to (a) establish the strongest classical/deep baseline under a subject-independent split, and (b) evaluate an initial GNN that models inter-channel structure explicitly.

II. Preprocessing and Split (No Leakage)

Subject-independent split: subjects 1–20 → train (22,724 windows); subjects 21–27 → test (8,212 windows). All preprocessing is fit on training data only: (1) median imputation, (2) StandardScaler, (3) SelectKBest (mutual information, top 100 of 128 features), (4) SMOTE oversampling on the selected training features only (→ 30,360 balanced training samples, 1,320/class). No test-set information enters any preprocessing step.

III. Baseline Models and Evaluation

Nine models were trained on the identical preprocessed/resampled training set and evaluated on the untouched test set, using both weighted and macro-averaged metrics plus training time.

Table 1 — Weighted metrics (primary comparison table)

Model	Accuracy	Precision	Recall	F1	Train time (s)
SVM (RBF)	0.7746	0.7836	0.7746	0.7750	100.88
Random Forest	0.7482	0.7536	0.7482	0.7462	20.34
LightGBM	0.7404	0.7593	0.7404	0.7423	34.30
XGBoost	0.7306	0.7473	0.7306	0.7302	61.06
MLP	0.6874	0.7082	0.6874	0.6865	44.42
k-NN (k=7)	0.6841	0.6930	0.6841	0.6853	0.00
1D-CNN	0.6248	0.6319	0.6248	0.6223	36.30
Logistic Regression	0.6231	0.6663	0.6231	0.6244	18.30
Decision Tree	0.4664	0.4687	0.4664	0.4607	9.58

Table 2 — Macro-averaged metrics (§6.1 full metric set)

Model	Macro-P	Macro-R	Macro-F1	ROC-AUC (macro, OvR)	PR-AUC (macro, OvR)
SVM (RBF)	0.7906	0.7879	**0.7842**	0.9814	0.8420
Random Forest	0.7572	0.7576	0.7514	0.9777	0.8065

Model	Macro-P	Macro-R	Macro-F 1	ROC-AUC (macro, OvR)	PR-AUC (macro, OvR)
LightGBM	0.7668	0.7467	0.7480	0.9790	0.8319
XGBoost	0.7601	0.7392	0.7400	0.9792	0.8269
MLP	0.7185	0.7106	0.7031	0.9720	0.7764
k-NN (k=7)	0.6876	0.7018	0.6903	0.9169	0.7024
Logistic Regression	0.6869	0.6606	0.6560	0.9575	0.7251
Decision Tree	0.4802	0.4755	0.4715	0.7255	0.2691
1D-CNN*	0.6331	0.6226	0.6198	0.9448	0.6565

*1D-CNN trains on raw normalized windows, not the selected-feature pipeline; its own train/test label encoding is used, so it is not directly leakage-comparable to the feature-based rows but is included for completeness.

Baseline to beat: SVM (RBF), Accuracy 0.7746, Macro-F1 0.7842. Per-class F1 for this model ranges from 0.56 (class 9) to 0.97 (class 20); the weakest classes (9, 10, 11, 16) are among those flagged in Task 1 as cross-exercise label collisions, consistent with that EDA finding.

IV. Proposed GNN

A. Graph construction. Nodes: all 32 channels (10 EMG + 22 glove). Node features: RMS, MAV, WL, VAR per channel per window (4-dim). Edges: pairwise Pearson correlation of per-window RMS values, computed on the training set only, thresholded at $|r| > 0.3$ $\rightarrow$ 184 of

992 possible directed edges. Edge weight = |correlation| (absolute value required because GCNConv's degree normalization needs non-negative weights; using signed correlation caused training collapse in an earlier iteration). Rationale: Task 1 established non-trivial

inter-channel correlation; this graph operationalizes that finding, but is one candidate construction, not a proven-optimal one.

B. Architecture. Input (32×4) → correlation graph → learnable edge-refinement MLP (1→8→1, sigmoid, scales each edge weight) → GCNConv(64)+BatchNorm+ReLU+Dropout(0.3) ×2 → GCNConv(64)+BatchNorm+ReLU → global mean pool → Linear(64→23). Total parameters: 10,544.

C. Explanation. Each of the 32 channels is a graph node carrying 4 window-level descriptors. Rather than fixing edge importance to the raw correlation value, a small learned gate adjusts each edge's weight during training, allowing the model to deviate from the initial correlation-based structure. Three GCN layers propagate information across this heterogeneous (EMG+glove) graph; mean pooling aggregates all 32 node embeddings into one window-level representation for classification.

D. Training. CrossEntropyLoss, Adam (lr 1e-3, weight decay 1e-5), ReduceLROnPlateau (patience 5, factor 0.5), 100 epochs, batch size 64. Class imbalance handled via random oversampling with replacement to the majority class count (SMOTE's linear interpolation does not apply cleanly to structured channel×feature node tensors). Training time: 763.6 s; best validation Macro-F1: 0.3925.

V. First Results — GNN vs. Baselines

Model	Macro-P	Macro-R	Macro-F1	ROC-AUC	PR-AUC
SVM (RBF)	0.7906	0.7879	0.7842	0.9814	0.8420
Random Forest	0.7572	0.7576	0.7514	0.9777	0.8065
LightGBM	0.7668	0.7467	0.7480	0.9790	0.8319
XGBoost	0.7601	0.7392	0.7400	0.9792	0.8269
MLP	0.7185	0.7106	0.7031	0.9720	0.7764
k-NN	0.6876	0.7018	0.6903	0.9169	0.7024

XGBoost	0.7601	0.7392	0.7400	0.9792	0.8269
Logistic Regression	0.6869	0.6606	0.6560	0.9575	0.7251
Decision Tree	0.4802	0.4755	0.4715	0.7255	0.2691
Proposed GNN (EMG+glove)	**0.4024**	**0.4374**	**0.3925**	0.8826	0.4230

The proposed GNN ranks last overall, below every classical baseline including Decision Tree. Test accuracy 0.3699, weighted-F1 0.3607. Per-class inspection shows severe over-prediction of certain classes (e.g., class 19: 770 predictions vs. 170 true support, precision 0.11) alongside reasonable performance on others (classes 21–23: F1 0.65–0.79), indicating the model has learned some structure but is far from competitive.

VI. Discussion

This is reported as a genuine first-iteration result, not adjusted after the fact. Plausible contributing factors, to be tested in Task 3's ablation: (1) a fixed 3-layer depth may over-smooth a 32-node graph; (2) the correlation threshold (0.3) and edge-weight scheme were not tuned; (3) random oversampling of structured node tensors may introduce less useful synthetic diversity than SMOTE provided the classical models; (4) a single GCN variant was tested — GAT or GraphSAGE, or an anatomical-adjacency graph, were not yet compared. The GNN is not required to beat the baselines at this stage; the requirement is a scientifically defensible next step, which Task 3 addresses via ablation.

VII. Conclusion

SVM (RBF) is the baseline to beat (Macro-F1 0.7842). The proposed correlation-graph GCN, in its first iteration, substantially underperforms all baselines (Macro-F1 0.3925). This gap, and the specific architectural/graph-construction choices most likely responsible for it, are the explicit subject of the Task 3 ablation study.

Deliverables: code/task2/Group04_NinaProDB1_task2_baselines.ipynb, Group04_NinaProDB1_task2_ProposedModel.ipynb; report/task2/Group04_NinaProDB1_task2_report.pdf.